

PHY111: Physics Laboratory I

End Semester Exam

Duration: 1 Hour

Total Marks:20

Write down all the necessary steps.

1. What is a random error? Explain it with an example. [1+1=2]
2. Write the formula for the standard deviation, explaining the meaning of all symbols used. What does standard deviation represent in the measurement of a variable? Explain it with an example. [1+1=2]
3. In a vernier calipers, 1 cm of the main scale is divided into 20 parts, and 39 main scale divisions match with 40 vernier scale divisions. Calculate the vernier constant. [3]
4. A screw gauge has a pitch of 0.5 mm. Its circular scale has 100 divisions. When measuring the diameter of a wire, the main scale reading is 2.5 mm and the circular scale reading is 73 divisions. If the zero error is -0.02 mm, calculate the correct diameter of the wire. [3]
5. A student measures the sides of a triangle as 3.0 cm, 4.0 cm, and 5.0 cm using a 10-cm scale in which 1 cm is divided into 10 equal parts. Assuming the maximum possible error in each measurement equals half the least count of the scale, calculate the maximum possible relative error in the measured perimeter. [3]
6. A student measures three physical quantities X, Y, and Z in order to calculate

$$A = \frac{XY^2}{Z^3}.$$

The measured values are given in the table below.

X (in cm)	9.87	10.45	11.02	10.11	9.98	10.56	10.72	9.65	10.89	9.05
Y (in mm)	2.612	2.843	2.517	2.901	2.688	2.734	2.559	2.813	2.469	3.374
Z (in cm)	6.12	5.98	6.45	6.77	6.29	6.51	6.04	6.88	6.23	6.13

Using the data provided, calculate the relative error in A.

[7]